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Exploring the indigenous medicinal plants and their health benefits in Patamda Region of Jharkhand: A comprehensive review

Kumari Chandrawati[a], Vijay Kant Pandey[b]* & Priyanka Jha[c]
[a]Department of Zoology, Netaji Subahs University, Jamshedpur, Jharkhand, India
[b]Department of Life Science, Netaji Subhas University, Jamshedpur, Jharkhand, India
[c]Project Girls High School, East Singhbhum, Jharkhand, India

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Abstract- Traditional plant-based medicine is crucial in Jharkhand due to the vast Indigenous flora supported by the forested lands. The present review presents ethnomedicinal practices in Patamda and discusses the people's traditional knowledge passed through generations. More than 160 medicinal plant species are used in curing fever, digestive disorders, and skin conditions. The researchers conducted an appropriate investigation involving 200 students from five schools to medicinally quantify these traditional remedies. The study's results are indispensable and medically verified by systematically analyzing the bioactive compounds, medicinal values, and people's socio-cultural uses.

Keywords: Ethnomedicine, bioactive compounds, indigenous healthcare, medicinal flora

INTRODUCTION

The boundary between traditional knowledge and modern scientific discourse is increasingly blurred in respect of eliciting the therapeutic potential of indigenous medicinal plants. In Patamda, Jharkhand, the local population has benefited from traditional plant-based medicine, which they have considered to be the primary healthcare resource for generations. Over time, they have developed complex systems of knowledge about how to identify, prepare, and apply the local flora to treat various diseases. However, the value of traditional knowledge goes far beyond the mere curiosity; it is a functioning pharmaceutical warehouse with minimal side effects and proven effectively in treating local diseases. In the previous project, conducted with 200 students from five Patamda schools, scientists documented these remedies and scientifically approved them assessing their efficiency and side effects while preserving the cultural heritage. Thus, the project under consideration contributes to the protection and dissemination of traditional knowledge.

*Corresponding author :
Phone : 8825290968
E-mail : pandeyvijay00@gmail.com

Medicinal Plants of Patamda: Therapeutic Applications and Bioactive Constituents

1. Babool (*Acacia nilotica* L.)

Babool is one of the most widely used medicinal plants in Patamda area, useful for the treatment of gastrointestinal diseases. This plant is rich in alkaloids and flavonoids, the metabolites responsible for its therapeutic activity. In folklore practice Babool fruits are administered twice a day continuously for 7 days to cure colic pain and indigestion. Phytochemical constituents such as sterols, flavonoids and tannins that accounting for therapeutic activity of Babool, while these secondary metabolites have

an antimicrobial and anti-inflammatory property due to which it is effective in treating the gut irritation. The analgesic effect comes from the alkaloids[1], while the anti-inflammatory action on the GI tract derives from flavonoids[2].

2. Bana Kundari (*Coccinia grandis*)

Coccinia grandis is an herbal medicine that is used to treat diabetes, sores, poor appetite, vomiting, and toxic substance ingestion. The compounds in the plant such as quercetin, and cephalandrine A, and assorted alkaloids contribute to its medicine properties. In the case of diabetes, the warm juice of leaves and roots is administrated at 2–3 teaspoons with the dosage dependent of the severity of the condition. For vomiting, leaf juice of 5–6 teaspoons can be administrated particularly for cases of poisoning. Additionally, fried leaves may help increase appetite. Externally, it can be used for abscesses as the plant has both antimicrobial properties and anti-inflammatory properties.[3]

3. Hadasinkuda (*Cissus quadrangularis* L.)

This species is of special concern for traditional treatment of bone fracture and the local name title in the ethnomedicinal usage. The medicinal concoction includes crushing the plant's stem and leaves along with one teaspoon of salt, five teaspoons of other ingredients to form a paste. This paste is pasted on affected part as plaster and held in position by means of bamboo sticks, to maintain its stability while healing. Its bioactive compounds such as flavonoids, steroids and triterpenoids have good osteoprotective effect and enhance bone regeneration thereby healing the fracture. It has also been scientifically confirmed that Cissus quadrangularis is osteogenic[4], corroborating traditional medicine attestation of its effectiveness in amelioration of skeletal afflictions.

4. Purani (*Boerhavia diffusa* L.)

As a response to the increasing menace of dog bite cases in rural India, Purani has been used for centuries by traditional healers in Patamda. The treatment consists of mashing a 4.5 in piece of the plant into a paste that is then taken by patients for 3 days in a row. The plant is rich in flavonoids, alkaloids, and rotenoids with antimicrobial and wound-healing potentials.[5] Although modern post-exposure prophylaxis is important for the prevention of rabies, the use of Purani does suggest a community's ability to manage wounds and prevent infection on site particularly in such areas where modern medical help may not be readily available.

5. Kanta Khira (*Amaranthus Spinosus* L.)

It is a folk medicine and is medications for colds, flu and fever. The curative aspect of the preparation is this ground-root paste, taken in five teaspoonful doses each day. Kanta Khira is rich in flavonoids and polyphenols, which explain its role as an anti-inflammatory and antipyretic.[6] The bioactive compounds combine to suppress inflammatory responses in airways, and immune response to viral infections. This ancient cure is a reflection of their deep knowledge of plant medicine.

6. Agnimantha (*Premna integrifolia* L.)

Snake bites are serious emergencies in rural areas, and Agnimantha is widely used as a traditional first-aid remedy. The treatment consists of orally taking macerated leaves, mixed with black pepper, and applying a paste of the roots to the site of the bite. Isoflavonoids and triterpenoids, active compounds in Agnimantha, may mitigate the effects of venom activity and inflammation.[7] The bioactive compound piperine in black pepper enhances absorption.[8] Agnimantha is important and should not be used as a substitute for antivenin, particularly in regions where medical care may be limited.

7. Ankranti (*Solanum virginianum* L.)

Therapeutic Indications Ankranti is effective in treating fever with opthalmological complaints such as redness and lacrimation. The dose of this preparation is 2–3 spoons twice a day for seven days after grinding the roots of plants with 21 black pepper grains to obtain a paste. The plant has alkaloidal compounds such as solamargine, solasonine, solanidine, and solasodine which show antipyretic, anti-inflammatory and antimicrobial effects.[9] The detail that black pepper was added to the powder for improving absorption and possibly exerting anti-inflammatory influence also represent the holistic approach of traditional medicine.

8. Ashwagandha (*Withania somnifera* L. Dunal)

Ashwagandha has a traditional use in the treatment of tuberculosis and malnutrition in children. The treatment for tuberculosis is the use of six grams of dried root powder, mixed with honey in a few ounces of water. This utilizes one of the withanolides in the plant, withaferin A, for its immune-modulating and antimicrobial properties.[10] Five grams with milk daily is recommended for nutritional support in pediatric patients with malnutrition. The adaptogens in the ashwagandha also help to modulate stress

and improve nutrient absorption, which will promote overall health, recovery, and concomitant care in both conditions.

9. Aamrataka (*Spondias pinnata* Lf. Kutz)

Aamrataka bark formulations prove to be a simple remedy for joint pain, especially in aged persons and at the extremes of ages. The plant has flavonoids, tannins, triterpenoids and polyphenols which together attributed for anti-inflammatory and analgesic activity.[11] Indigenous healers make a liniment or paste from the bark which is rubbed on painful, arthritic joints. Tannins are thought to have astringent effects and prevent or reduce swelling while Polyphenols are believed to be antioxidants that slow down degenerative joint steps.

10. Vajrakantakah (*Euphorbia antiquorum* Wall.)

The plant treats swellings of the throat by applying its milky latex to the affected area once daily for one week. The latex also contains diterpene, flavonoids and triterpenes with anti–inflammatory action.[12] The classical route of administration results in the direct contact with inflamed tissues and modulation exerted by its bioactive components on site. But near practitioners should guard caution since Euphorbia latex is believed to be an irritant which emphasizes the need for correct indigenous knowledge when using medicinal plants.

11. Guluchi (*Tinospora cordifolia*)

The essential medicinal plant is *Tinospora cordifolia*, which is used in the treatment of urogenital disorders, helminthiasis, gonorrhea and respiratory tract problems. The stem is the part of the plant that is primarily used, boiled or cold water extracts may vary depending on condition. For respiratory relief, it is mixed with *Justicia adhatoda* or *Ocimum tenuiflorum*. Containing alkaloids and glycosides, it has immunomodulatory, antimicrobial and anti-inflammatory activities.[13]

12. Patal Garuda (*Rauvolfia serpentina* L. Benth Ex.curz)

Ear-related ailments including sores and aches find treatment through direct administration of leaf juice extracted from Patal Garuda. The plant contains reserpine, ajmaline, and ajmalicine: alkaloid compounds with demonstrated analgesic and antimicrobial properties.[14] The traditional practice of direct ear application ensures high local concentration of bioactive compounds at the site of infection or inflammation, maximizing therapeutic efficacy while minimizing systemic exposure.

Table 1: Indigenous Medicinal Plants of Patamda Region: Traditional Uses and Bioactive Compounds

Plant Name	Primary Traditional Uses	Key Bioactive Compounds
Babool (*Acacia nilotica* L.)	Colic pain, indigestion	Alkaloids, Flavanoids
Bana Kundari (*Coccinia grandis* G.)	Diarrhea, abscesses, vomiting, appetite loss	Quercetin, Cephalandrine A, Alkaloids
Hadasinkuda (*Cissus quadrangularis* L.)	Bone fracture healing	Flavonoids, Steroids, Triterpenoids
Purani (*Boerhavia diffusa* L.)	Dof bite infection, wound healing	Flavonoids, Alkaloids, Rotenoids
Kanta Khira (*Amaranthus spinosus* L.)	Cold, fever, cough	Flavonoids, Polyphenols
Agnimantha (*Premna integrifolia* L.)	Snake bite treatment	Isoflavonoids Neoflavonoids, Triterpenoids
Ankranti (*Solanum virginianum* L.)	Fever, eye redness, excessive tearing	Solamargine, Solasonine, Solanidine
Aswagandha (*Withania somnifera* L. Dunal)	Tuberculosis, underweight children	Withaferin A, Withanone
Aamrataka (*Spondias pinnata* (L.f.) Kutz)	Painful joints, Arthritis	Flavonoids, Tannins, Triterpenoids, Polyphenols
Vajrakamozuku (*Gomphrena celosioides* Mart.)	Throat swelling	Diterpenes, Flavonoids, Triterpenoids
Guluchi (*Tinospora cordifolia*)	Respiratory difficulties, gonorrhea, helminthiasis	Alkaloids, Diterpenoid lactones, Glycosides, Steroids
Patal Garuda (*Rauvolfia serpentina* (L.) Benth. Ex Kurz)	Ear sores, earache	Reserpine, Ajmaline, Ajmalicine

CONCLUSION

The ethnomedicinal practices of Patamda hold rich therapeutic knowledge passed down through generations. A student-led research project scientifically validated many traditional remedies, revealing the community's deep understanding of medicinal plants. This fusion of tradition and science opens doors for new treatments and sustainable healthcare. However, further research is needed-phytochemical studies, clinical trials, and conservation efforts are vital. Planting medicinal species near schools and hospitals can aid access and awareness. Integrating traditional wisdom with science offers innovative, culturally respectful healthcare solutions.

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